



基于可穿戴惯性测量设备的老年人步态关键特征提取和多因素分析研究

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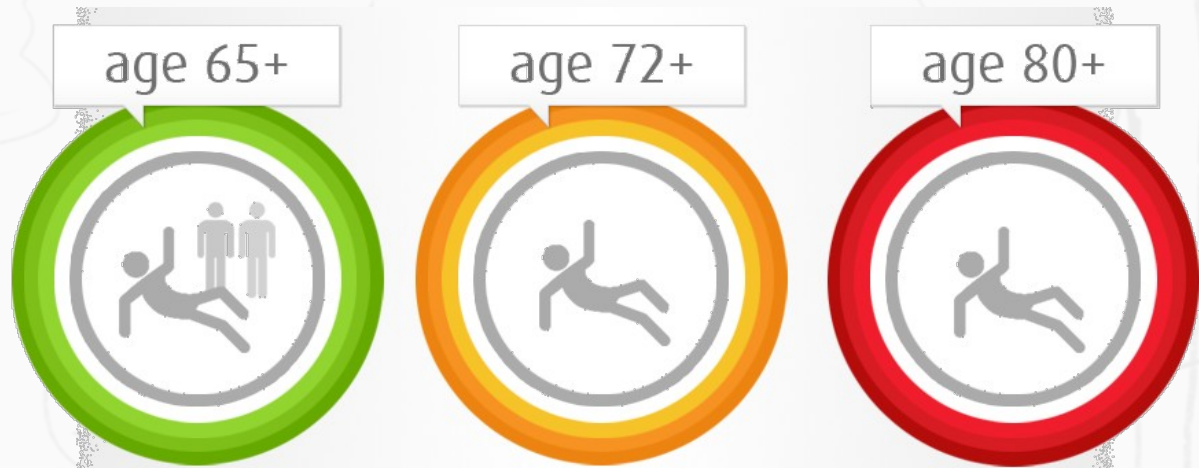
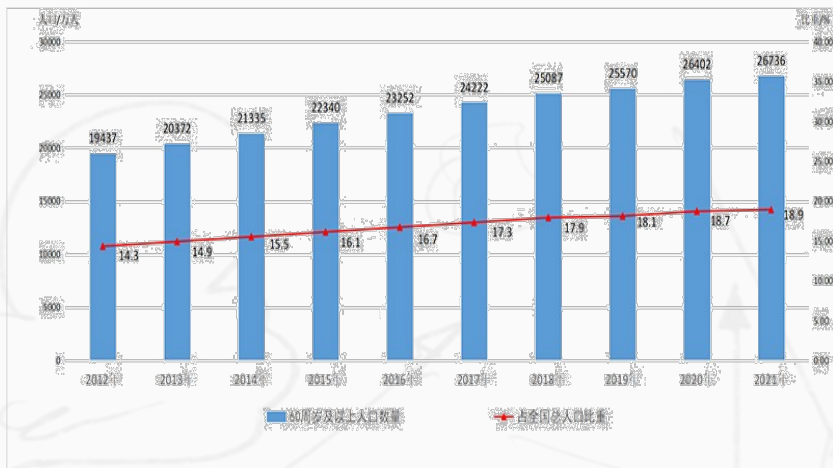




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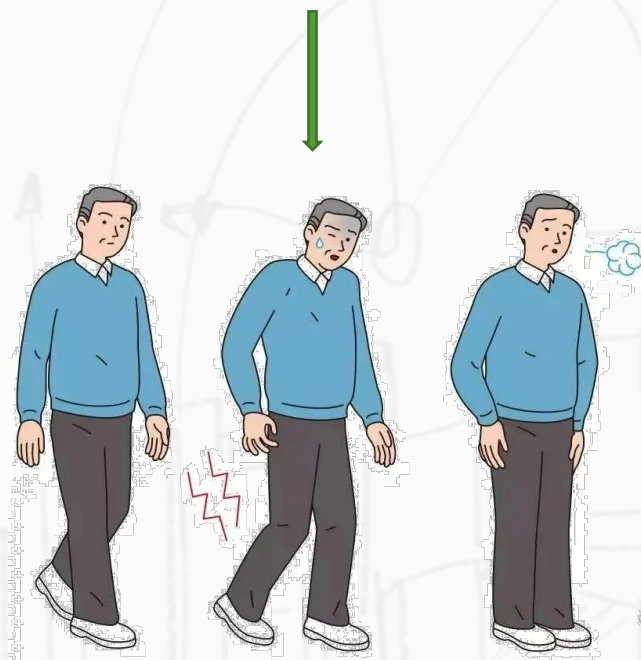
研究背景及意义

Research background and significance



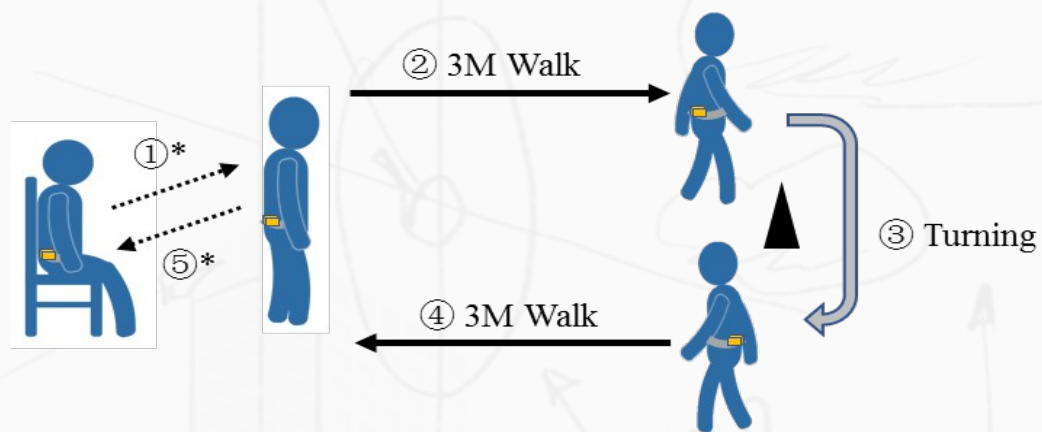
步态与平衡障碍

- 2035 年左右，60 岁及以上老年人口将突破 4 亿，在总人口中的占比将超过 **30%**，进入重度老龄化阶段。
- **步态和平衡障碍** 的发生率随年龄增长而增加，在 60 至 69 岁之间为 10%，在 80 岁以上的人中超过 60%。
- **步态和平衡障碍** 容易导致跌倒的发生，而跌倒又是老年人意外伤害死因的首位原因。



eg : 3 Meter Timed Up and Go (3M-TUG)

一种常用的测量功能移动性和动态平衡的方法，具有耗时短、操作简单、可靠性高的特点。



* Note: ① Sit-to-stand; ⑤ Stand-to-sit Accelerometer & gyroscope

面临的挑战

参数准确性

大部分参数提取后直接使用，没有经过检验分析；

切分位点不统一

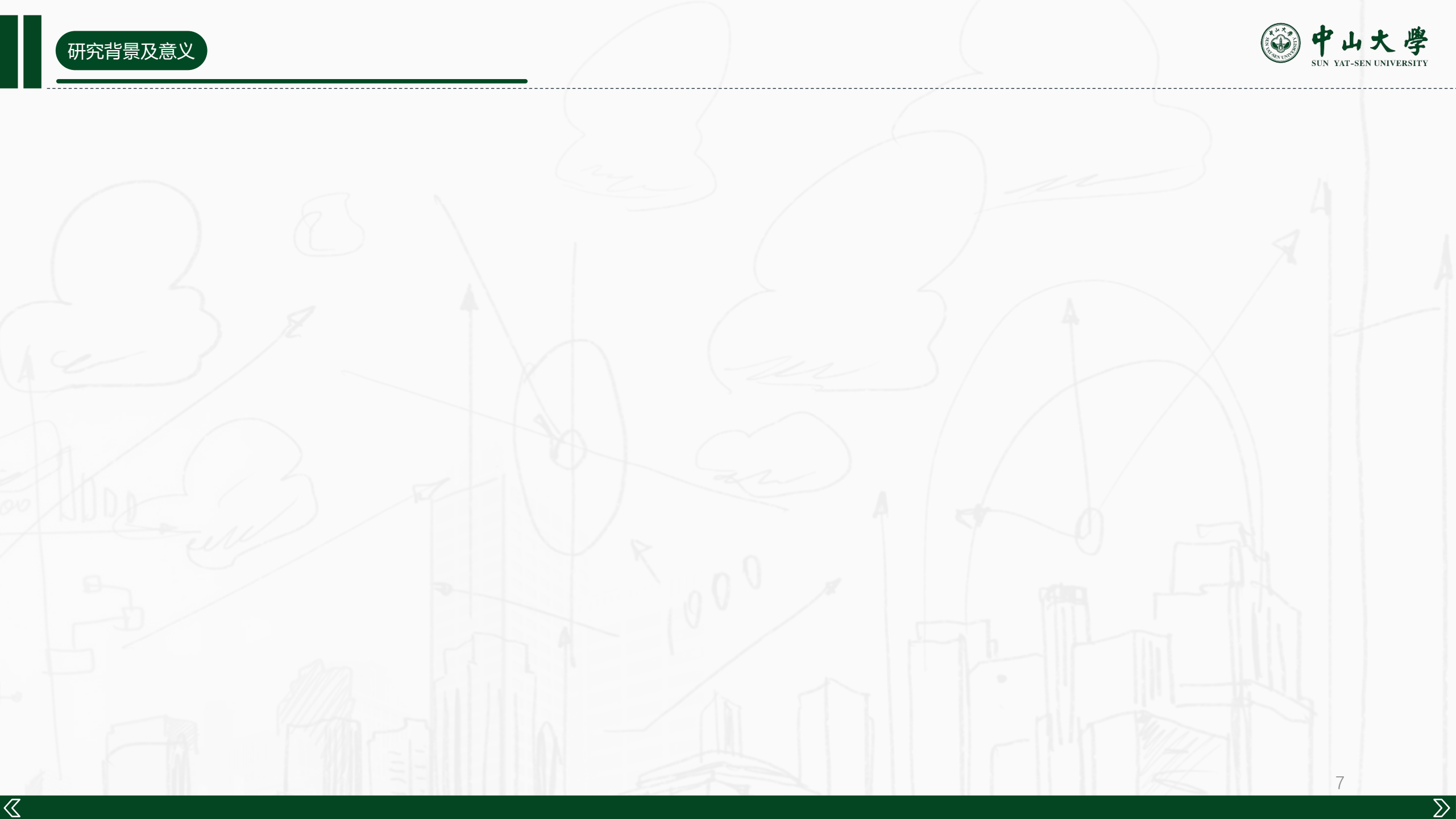
例如，Yu-Cheng Hsu 等人 [5] 和 Jungyeon Choi 等人 [6] 在切分采集 TUG 过程的传感器数据时，使用的切分标准不同。

跌倒风险的评定

对跌倒风险的评定标准不明确使得各研究结果不一致；

参数多样性

现有的步态分析系统及纳入的步态参数多种多样，过于繁琐。





02

研究框架与思路

Research framework

缺失值检查

数据低通滤波去噪

TUG 时相划分

从Kinect数据中提取特征进行步态参数一致分析

- 归一化处理
- 构建模型(极限学习算法)

特征提取

- 提取步态时间参数：TUG完成时间、步速、步频、转身时间、坐到站持续时间，行走过程中加速度和角速度的变异系数、均值等
- 提取步态空间参数：步长
- 基于频谱的步态参数：振幅的均方根、最大值、最小值、范围
- 人口学特征：年龄、学历、疾病史、吸烟、喝酒等

机器学习

- 结合IMU和Kinect数据构建深度学习的卷积神经网络CNN

统计检验

- 聚类分析
- t检验
- Mann-Whitney U检验

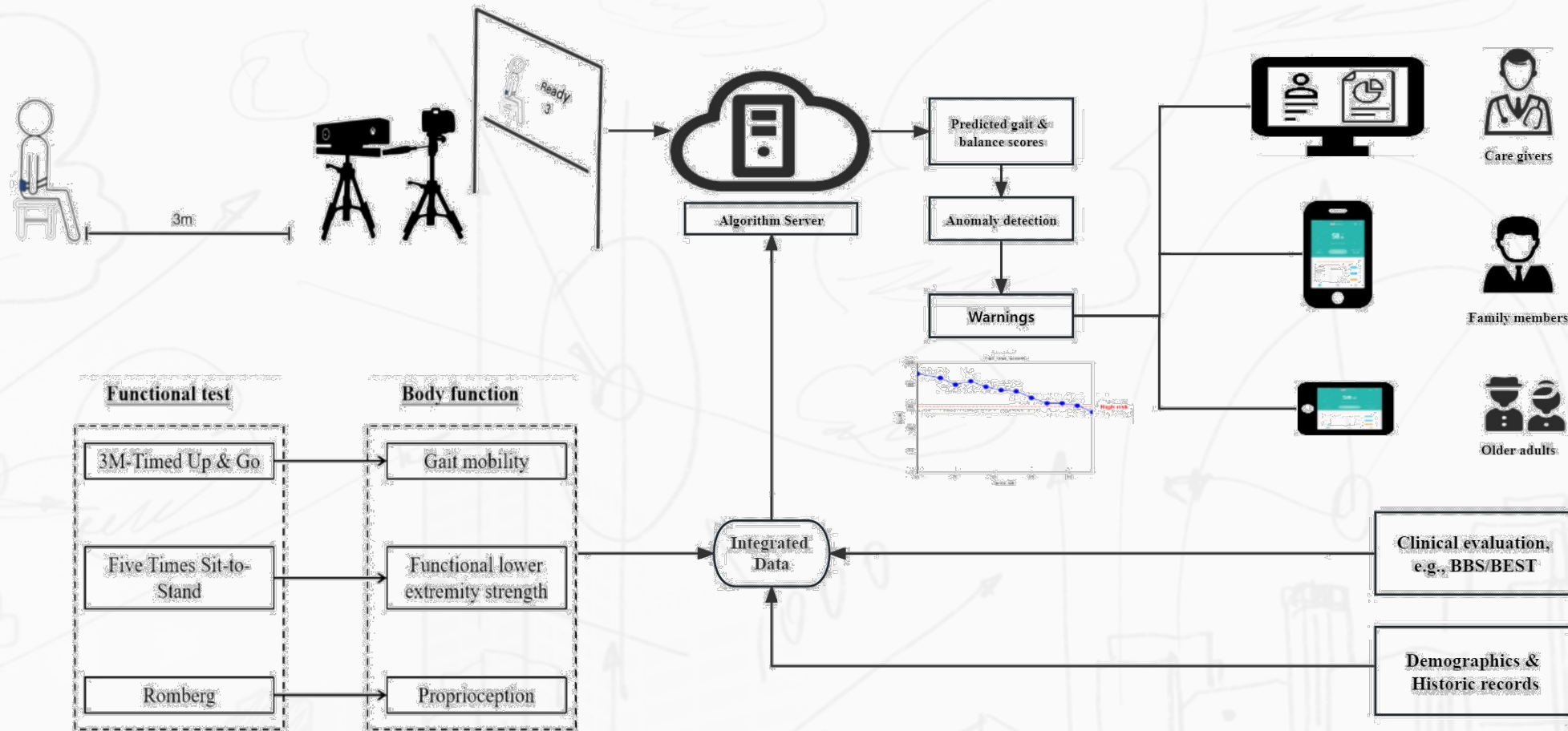


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初步成果

Preliminary results

An integrated sensor-based intelligent system prototype for fall risk assessment in older adults: A feasibility study



Schematic diagram of the proposed fall-risk-assessment system prototype.

An integrated sensor-based intelligent system prototype for fall risk assessment in older adults: A feasibility study



Illustration of the three clinical standard tests (a) 3M-TUG test, (b) Romberg test, (c) FTSS test

人口统计学特征分析

- 使用**卡方检验**来分析跌倒风险和人口统计学特征之间的关联。所有的测试都是双侧检验， α 值设定为 0.05。计算了人口统计学变量的频率。

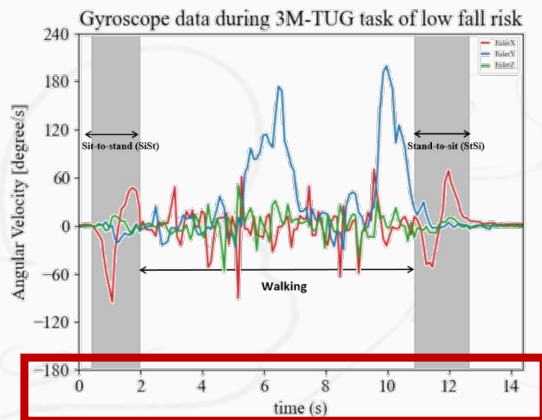
在以下变量中，高风险组和低风险组之间存在着明显的差异："心血管疾病"、"降压药"、"不吸烟"、"中风"、"使用助行器"和"跌倒史"。

The Significant general characteristics of the participants.

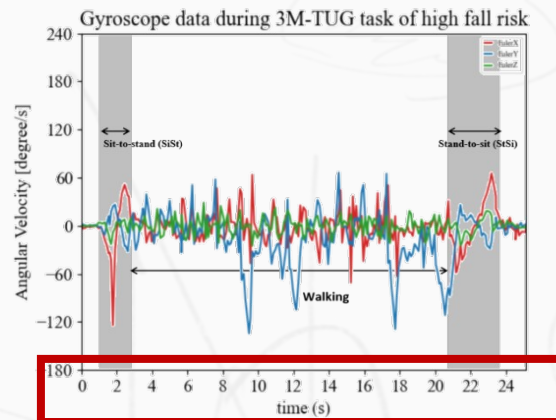
Parameter	Low risk fallers	High risk fallers	p Value
Participants	30	26	
Fall history (Y/N)	2/28	13/13	0.000**
Gender (F/M)	14/16	6/20	0.066
Cardiovascular disease (Y/N)	10/20	19/7	0.003**
Osteoporosis(Y/N)	10/20	12/14	0.327
Antihypertensive medication (Y/N)	14/16	21/5	0.009**
Antihyperlipidemic drugs (Y/N)	9/21	7/19	0.799
Arthralgia (Y/N)	11/19	4/22	0.073
No smoking (Y/N)	24/6	13/13	0.018*
Stroke (Y/N)	11/19	17/9	0.032*
Usage of walking assistance (Y/N)	0/30	14/12	0.000**

* $p < 0.05$, ** $p < 0.01$

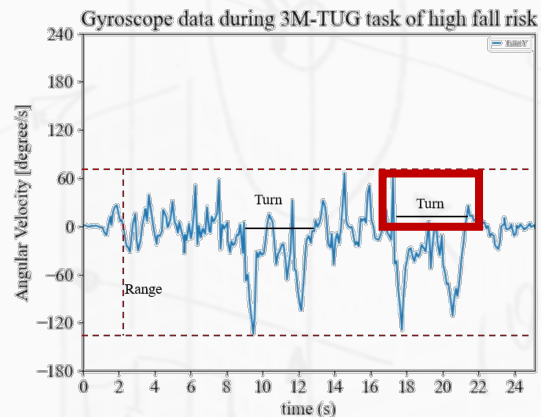
IMU 可行性分析



(a)



(b)



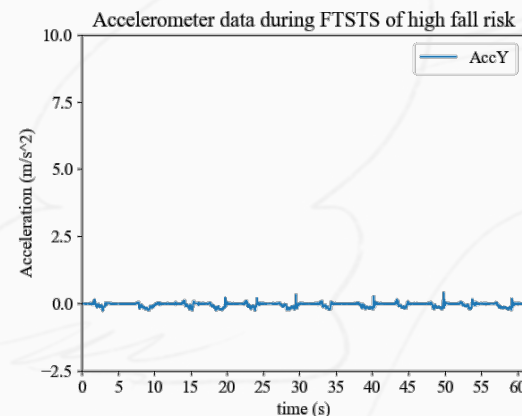
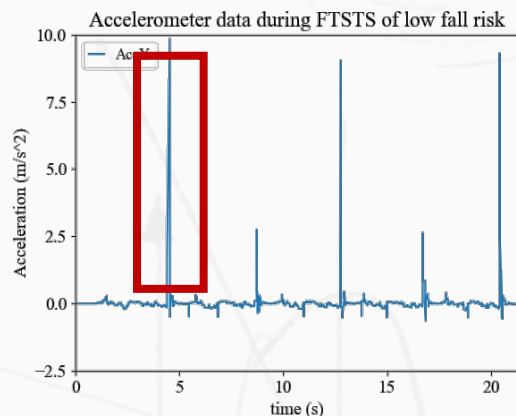
(d)

Gyroscope data during 3M-TUG task of participants with low and high fall risk (a) low fall risk of the 3-axis; (b) high fall risk of the 3-axis; (c) low fall risk of the y-axis; (d) high fall risk of the y-axis.

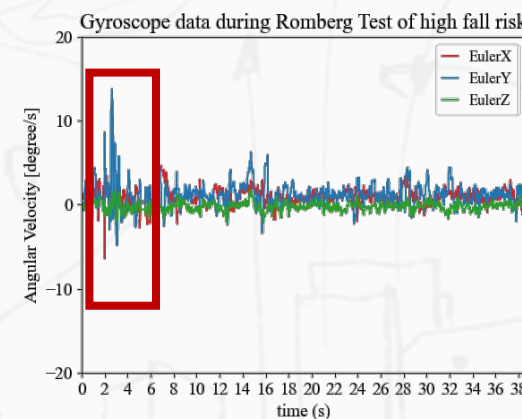
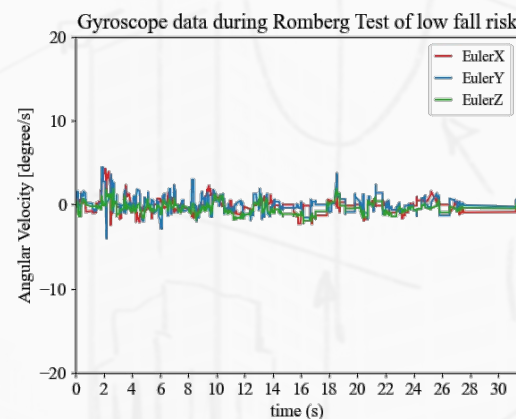
高风险跌倒者

- 更长 THG 时间
- 角速度变化幅度稍大
- 转身状态不明显

IMU 可行性分析



Accelerometer data during Five Times Sit-to-Stand Tests from low-risk and high-risk fallers



Gyroscope data during Romberg Test of participants with low and high fall risk of the 3-axis

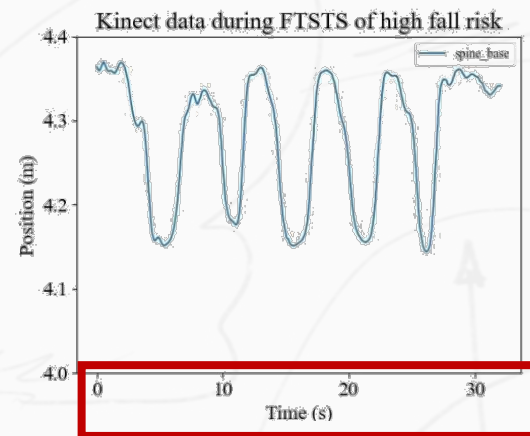
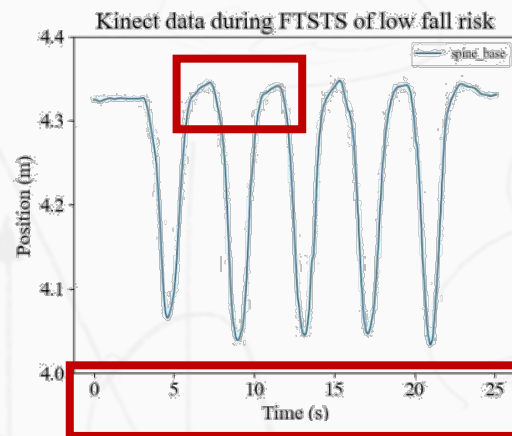
高风险跌倒者

- 每次起坐的加速度更快
- Romberg 的角速度变化更大

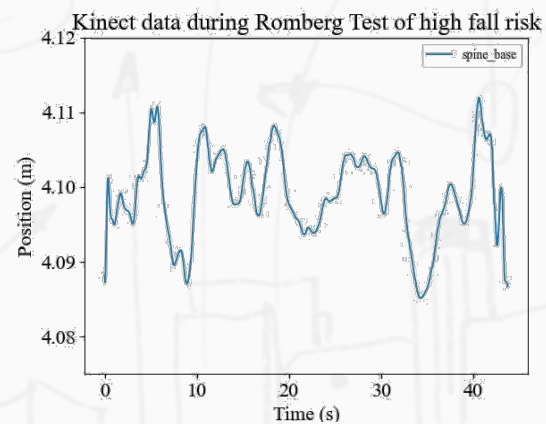
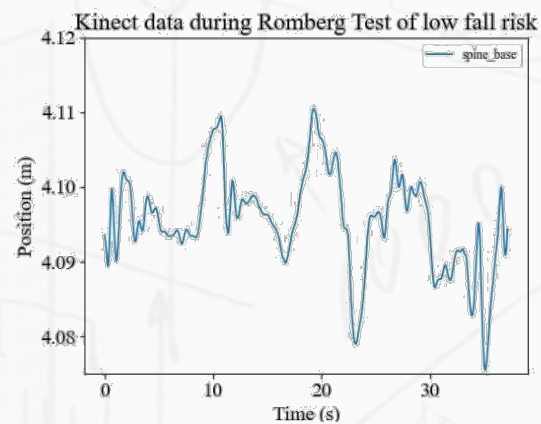
深度相机可行性分析

Kinect data during 3M-TUG task of participants with low and high fall risk of the z-axis with ankle diff

深度相机可行性分析



Kinect data during Five Times Sit-to-Stand Test of participants with low and high fall risk of the z-axis with spine base

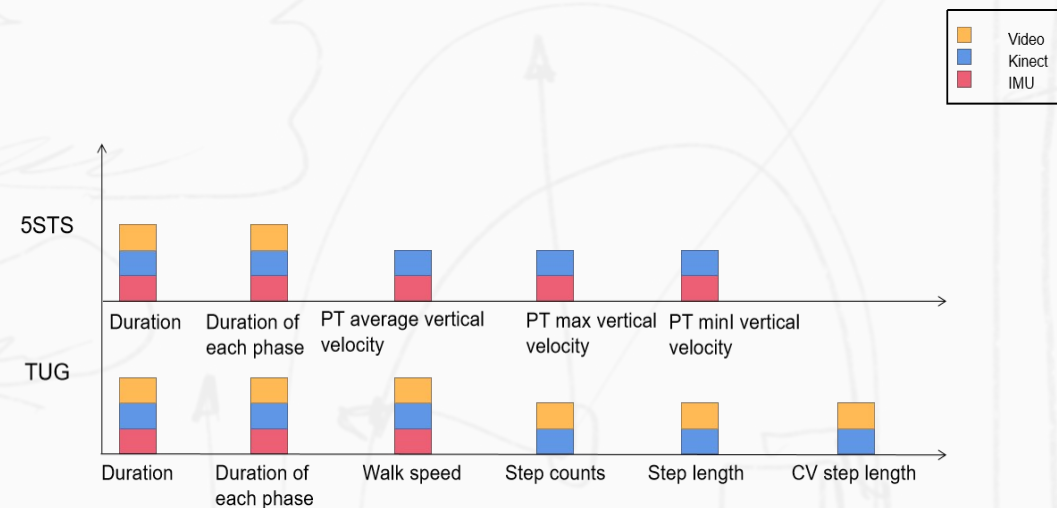


Kinect data during Romberg Test of participants with low and high fall risk of the z-axis with spine base

Comparison and validation of key features extracted from different sensors

Summary of features of TUG.

Device	TUG Features
IMU	Duration , Maximum, Minimum, Range, RMS, Median, Max jerk, Mean jerk, CV, duration of each phase , Walk speed
Kinect	Duration , Legs angle, Left knee angle, Walk speed , stride mean, cadence, Right knee angle, step counts , Step length , Step width, duration of each phase , the width of turn, speed to get up, speed to sit down, turn POD, walk POD, get up POD, sit down POD, CV step length , Euler Angle, Euclidean Distance, Body Lean Angle, Left arm Euler Angle, Euclidean distance between the left shoulder and left hand, Left leg Euler Angle, Euclidean distance between the left hip and left foot, Right arm Euler Angle, Euclidean distance between the right shoulder and right hand, Right leg Euler Angle, Euclidean distance between the right hip and right foot, Torso Euler Angle, Euclidean distance between the spine base and head
Video	Duration , duration of each phase , step length , CV Step length , Step counts , Walk speed



The consistent features extracted from IMU, Kinect, or Video data

Note. POD: Percentage of duration (subtask duration / TUG duration); RMS: root mean square; CV: coefficient of variation; Jerk: the rate of change in acceleration; COM: Center-of-Mass; Legs angle: the angle between the two legs is defined as the angle between the two vectors of the line between the joints 0 and 17 and the line between 0 and 13; Knee angle: it is defined as the angle between the thigh and the calf, for example, the Right knee angle is the angle between the two vectors 17 and 16 of the joint and 17 and 18 of the joint; **PT: postural transition**; Left arm Euler Angle: it is defined as the angle between left shoulder and left wrist; Left leg Euler Angle: it is defined as the angle between the left hip and left ankle; Torso Euler Angle: it is defined as the angle between the spine base and neck; Euclidean distance: it is defined as the distance in x and y axis vectors; Body lean angle: it is defined as the angle relative to the floor with the torso as a reference; Bolded features: it is defined as the common features that sensors can extract.



04

后续工作计划

Future works

深度学习

- **自动特征学习**：深度学习通过神经网络的多层结构，可以自动地学习出更加抽象和高级的特征表示。传统的机器学习方法通常需要手工设计和选择特征，这个过程依赖于专业知识和经验，而深度学习能够通过大规模数据的训练，**自动地从原始数据中学习到的最优的特征表示**。
- **高度非线性**：深度学习使用的神经网络模型通常具有多层非线性变换，这使得深度学习模型能够更好地**捕捉数据中的复杂非线性关系**。相比之下，传统的机器学习方法如逻辑回归、支持向量机等往往只能建模线性或浅层非线性关系，无法很好地处理高度非线性的特征提取任务。
- 在小样本数据上进行深度学习是一项挑战，深度学习模型通常需要大量的数据才能学习到有效的特征和模式。然而，还是有一些方法可以在小样本数据上进行深度学习，数据增强、生成噪声、L1 正则化等。

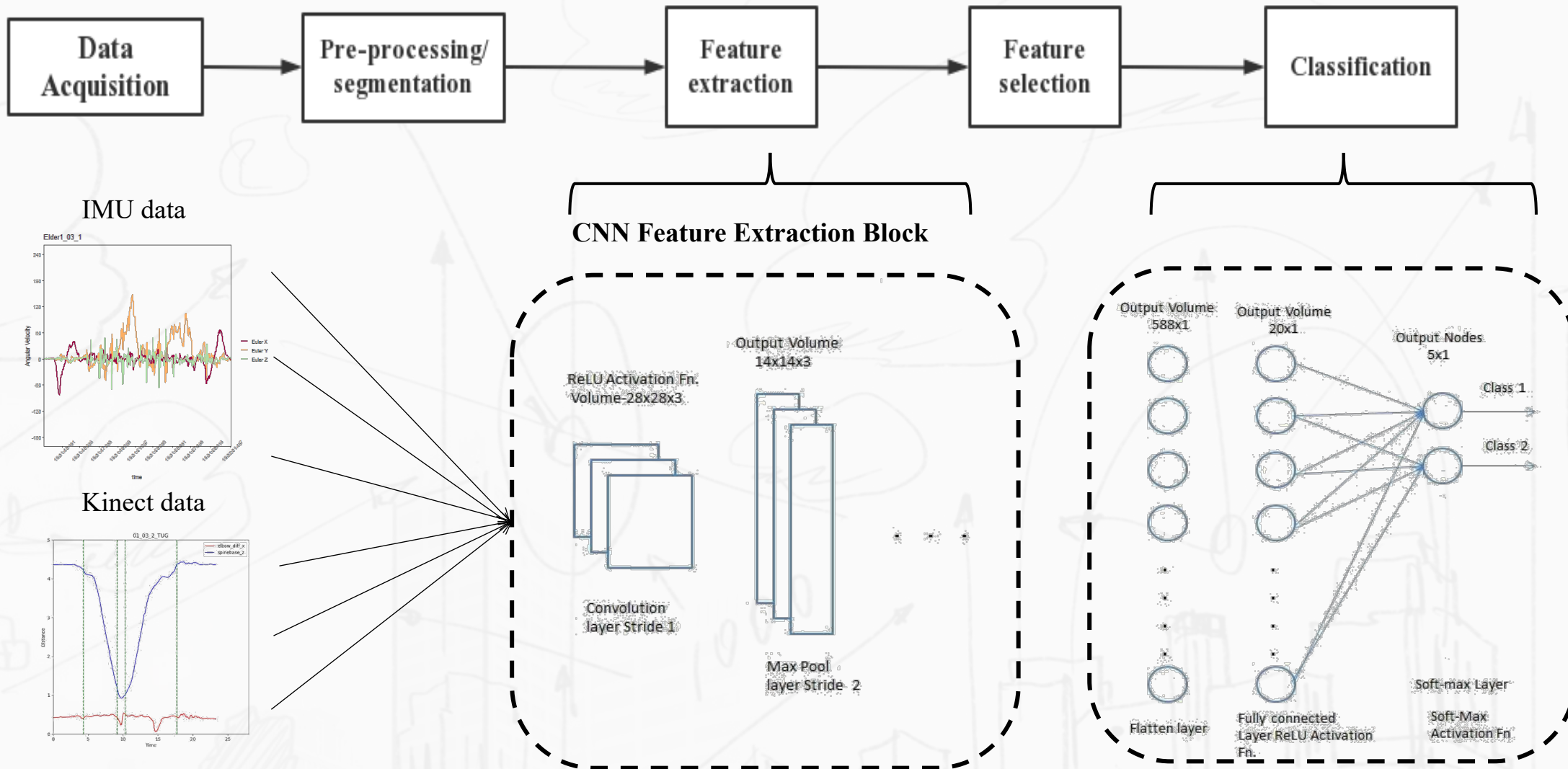
构建简单的 CNN 结构

CNN 模型适用于处理已知类网格拓扑的 1D、2D 或 3D 数据 [31]。通过对输入数据进行卷积操作，该网络能够从大型数据集中学习特征。通常，网络由卷积层、池化层和归一化层组成，并在这些层之间共享一组过滤器和权重。

卷积层输出从原始输入数据自动获取的特征图。池化层被用来减少表征的大小，使卷积层的输出更加鲁棒 [29]，[30]。CNN 模型通常使用两种池化层：最大池化和平均池化。所有的卷积层和池化层都有激活函数（如 Sigmoid, Tanh, ReLU, Leaky ReLU）来计算神经元的权值，并添加一个偏置，决定是否激活神经元 [32]。

LSTM 网络有利于处理时间序列数据，其中顺序很重要，如步态数据序列。从本质上讲，它们利用递归性，利用网络上前一个向前传递的信息。

结合 TUG 的 IMU 数据和 Kinect 数据，对步态异常做二分类变量，做数据增强后用 CNN 模型。



Convolutional neural network to extract and classify gait features from depth camera and wearable inertial measurement unit with accelerometer and gyroscope sensors.



感谢各位导师悉心指导

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